

CS471 – Block 2 Worksheet Solutions

Lessons 11-15: Markov Decision Processes and Reinforcement Learning

Block 2 Equations Reference

- **Discounted Return:** *Computes the cumulative reward from a sequence of steps, where the discount factor $\gamma \in [0, 1]$ bounds the value of infinite future returns.*

$$G = r_1 + \gamma r_2 + \gamma^2 r_3 + \dots$$

- **Value Iteration (Bellman Update):** *Calculates the maximum expected value of state s by checking all actions a and summing over all possible next states s' , weighted by transition probability $P(s'|s, a)$ and discount factor γ .*

$$V_{k+1}(s) = \max_a \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V_k(s')]$$

- **Policy Evaluation:** *Computes the expected value of state s when strictly following a specific, fixed policy π .*

$$V^\pi(s) = \sum_{s'} P(s'|s, \pi(s)) [R(s, \pi(s), s') + \gamma V^\pi(s')]$$

- **Q-Learning Update:** *Updates the value of taking action a in state s based on the learning rate α , immediate reward r , and the discounted max expected value of the resulting next state s' .*

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]$$

Part I: Guided Practice

Lesson 11. MDP Formulation and Markov Property

An autonomous cyber-defense agent routes analyzed data packets between secured enclave servers. A student proposes the following MDP:

- **State:** the packet's current server node.
 - **Actions:** forward, inspect, drop, or encrypt.
 - **Transition model:** a forward action always successfully reaches the intended neighboring server.
 - **Reward:** +5 when a packet reaches the destination enclave; 0 otherwise.
- (a) Identify two pieces of information that should be added to the state to better satisfy the Markov property. Briefly explain why each matters.

Solution

Possible answers include **encryption status** and **threat level** of the current server. Encryption status affects future inspection capabilities and rewards, while threat level affects the probability of packet dropping on subsequent forward actions. (Other defensible state information receives credit if its relevance to future outcomes is explained).

- (b) Give one realistic reason a forward action could have more than one outcome. How should the transition model represent this uncertainty?

Solution

A forward action might fail due to network congestion, hardware failure, or adversarial interference. The transition model should assign probabilities to the possible outcomes (e.g., $P(\text{Success}) = 0.95$, $P(\text{Dropped}) = 0.05$) rather than assume a single deterministic result.

Lesson 12. Policies, Values, and Return

An autonomous surveillance UAV follows the fixed policy shown below. Assume transitions are deterministic.

State	Policy Action	Immediate Reward	Next State
S_1	Scan	+2	S_2
S_2	Transmit	+6	Terminal

Use a discount factor of $\gamma = 0.5$.

- (a) Starting in state S_1 and following the policy, compute the discounted return.

Solution

$$\begin{aligned}\text{Return} &= r_1 + \gamma(r_2) \\ &= 2 + 0.5(6) \\ &= \mathbf{5}\end{aligned}$$

- (b) What does the state value $V^\pi(S_1)$ represent in this situation?

Solution

$V^\pi(S_1)$ is the expected discounted cumulative reward when starting in S_1 and then strictly following policy π . Here, $V^\pi(S_1) = 5$.

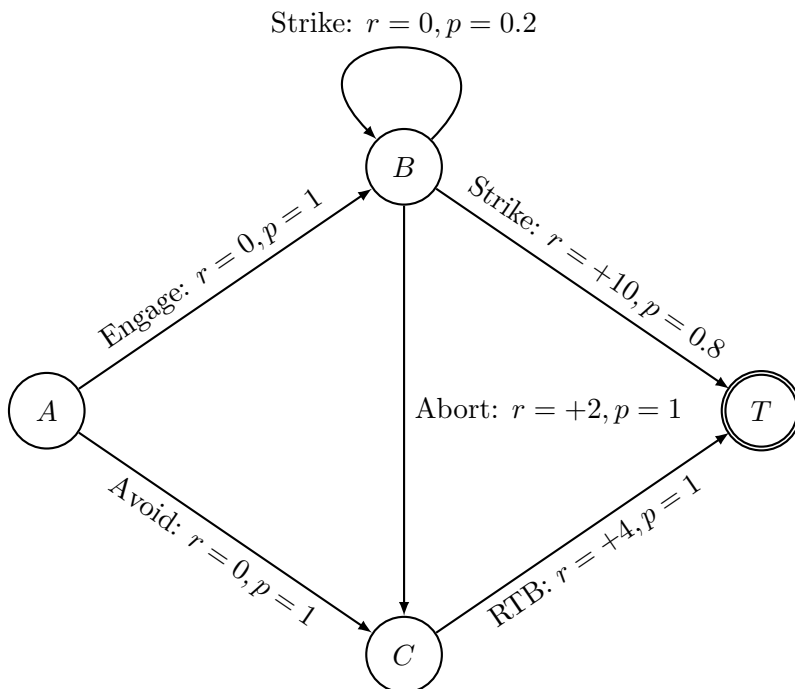
- (c) Suppose a different policy chooses “Loiter” at S_1 , receiving an immediate reward of +4 and terminating. Which policy gives the larger expected return from S_1 ?

Solution

The original policy gives a return of **5**. The alternative policy gives a return of **4**. Therefore, the **original policy** is better from S_1 .

Lesson 13: Value Iteration

An autonomous surface vessel uses the MDP below. Let $\gamma = 0.5$, $V(\text{Terminal}) = 0$, and $V_0(A) = V_0(B) = V_0(C) = 0$. Use synchronous updates.



- (a) Complete the first Value Iteration sweep to find $V_1(A)$, $V_1(B)$, and $V_1(C)$.

Solution

For B , the stochastic Strike action yields $0.8(10) + 0.2(0) = 8$. The Abort action yields $2 + 0.5(0) = 2$.

$$V_1(B) = \max(8, 2) = \mathbf{8}$$

For C , RTB yields $4 + 0.5(0) = 4$.

$$V_1(C) = \mathbf{4}$$

For A , Engage yields $0 + 0.5(0) = 0$. Avoid yields $0 + 0.5(0) = 0$.

$$V_1(A) = \max(0, 0) = \mathbf{0}$$

- (b) Using V_1 , compute only $V_2(A)$. Show the comparison between the available actions.

Solution

$$\text{Engage} = 0 + 0.5(V_1(B)) = 0 + 0.5(8) = 4$$

$$\text{Avoid} = 0 + 0.5(V_1(C)) = 0 + 0.5(4) = 2$$

$$V_2(A) = \max(4, 2) = 4$$

(c) After additional sweeps, suppose the values are: $V_k(A) = 3.80$, $V_k(B) = 8.12$, $V_k(C) = 4.00$, and $V_{k+1}(A) = 3.82$, $V_{k+1}(B) = 8.15$, $V_{k+1}(C) = 4.00$.

(i) Using the V_{k+1} values, determine the greedy policy action at state A . Show the action comparison.

Solution

$$\text{Engage} = 0 + 0.5(V_{k+1}(B)) = 0 + 0.5(8.15) = 4.075$$

$$\text{Avoid} = 0 + 0.5(V_{k+1}(C)) = 0 + 0.5(4.00) = 2.00$$

Therefore, the greedy action is **Engage**.

(ii) If the convergence tolerance is $\theta = 0.05$, has Value Iteration converged? Briefly justify.

Solution

Yes. The largest change across all states is:

$$\begin{aligned} \max(|3.82 - 3.80|, |8.15 - 8.12|, |4.00 - 4.00|) &= \max(0.02, 0.03, 0) \\ &= 0.03 \end{aligned}$$

Since $0.03 < 0.05$, the convergence criterion is satisfied.

Lesson 14: Policy Iteration

An autonomous logistics agent is being optimized using Policy Iteration. For the current policy, $V^\pi(Y) = 8$ and $V^\pi(Z) = -4$. Use a discount factor of $\gamma = 0.5$. The relevant transitions are:

State	Action	Next State	Reward
X	Option 1	Y	-2
X	Option 2	Z	-2
W	Option 3	Y	-2
W	Option 4	Z	-2

The current policy selects **Option 2** at X and **Option 4** at W .

- (a) Under the current policy, compute $V^\pi(X)$.

Solution

The current policy at X selects Option 2 (transitioning to Z).

$$\begin{aligned} V^\pi(X) &= -2 + 0.5(V^\pi(Z)) \\ &= -2 + 0.5(-4) \\ &= -4 \end{aligned}$$

- (b) Perform policy improvement.

- (i) At state X , compare the available actions and give $\pi'(X)$.

Solution

$$\begin{aligned} \text{Option 1 (to } Y) &= -2 + 0.5(8) = 2 \\ \text{Option 2 (to } Z) &= -2 + 0.5(-4) = -4 \end{aligned}$$

$$\pi'(X) = \text{Option 1.}$$

- (ii) At state W , compare the available actions and give $\pi'(W)$.

Solution

$$\begin{aligned} \text{Option 3 (to } Y) &= -2 + 0.5(8) = 2 \\ \text{Option 4 (to } Z) &= -2 + 0.5(-4) = -4 \end{aligned}$$

$$\pi'(W) = \text{Option 3.}$$

- (c) Based on the improved policy, what should Policy Iteration do next? Briefly justify your answer.

Solution

Both policy actions changed, meaning the policy is not stable. Policy Iteration should next **evaluate** the newly improved policy π' to compute the updated state values.

Lesson 15: Q-Learning from Experience

A network-security AI has the following Q-values:

$$Q(N_1, \text{Scan}) = 0, Q(N_2, \text{Patch}) = 0, Q(N_2, \text{Ignore}) = 2.$$

Use $\alpha = 0.5$ and $\gamma = 0.5$. State T is terminal.

Process the following experiences in order. Use any updated Q-values in the next experience.

- (a) The agent experiences $(N_2, \text{Patch}, 8, T)$. Compute the updated value of $Q(N_2, \text{Patch})$.

Solution

Since T is terminal, $\max_{a'} Q(T, a') = 0$.

$$\begin{aligned} Q(N_2, \text{Patch}) &= Q(N_2, \text{Patch}) + \alpha[r + \gamma(0) - Q(N_2, \text{Patch})] \\ &= 0 + 0.5[8 + 0 - 0] \\ &= 4 \end{aligned}$$

- (b) The agent then experiences $(N_1, \text{Scan}, 0, N_2)$. Compute the updated value of $Q(N_1, \text{Scan})$.

Solution

After part (a), $\max_{a'} Q(N_2, a') = \max(4, 2) = 4$.

$$\begin{aligned} Q(N_1, \text{Scan}) &= Q(N_1, \text{Scan}) + \alpha[r + \gamma \max_{a'} Q(N_2, a') - Q(N_1, \text{Scan})] \\ &= 0 + 0.5[0 + 0.5(4) - 0] \\ &= 0.5[2] \\ &= 1 \end{aligned}$$

- (c) Suppose the experience in part (b) had occurred before the experience in part (a). Would the resulting update to $Q(N_1, \text{Scan})$ be smaller, the same, or larger? Briefly justify using the Q-values at state N_2 .

Solution

The update would be **smaller**. Before part (a), the best Q-value at N_2 was $\max(0, 2) = 2$. If part (b) occurred first, the update would be:

$$\begin{aligned} Q(N_1, \text{Scan}) &= 0 + 0.5[0 + 0.5(2) - 0] \\ &= 0.5 \end{aligned}$$

Because part (a) increased the maximum expected future value at N_2 from 2 to 4, the subsequent update in part (b) is larger ($1 > 0.5$).

Part II: Independent Student Practice

Practice 1: MDP Formulation and Markov Property

An orbital satellite navigates between orbital slots to optimize communication coverage.

- **State:** The satellite's current orbital slot.
 - **Actions:** Thrust-Prograde, Thrust-Retrograde, or Sleep.
 - **Transition model:** A Thrust action always successfully moves the satellite to the adjacent slot.
 - **Reward:** +10 when reaching the target comms slot; 0 otherwise.
- (a) Identify two physical or system variables that should be added to the state to better satisfy the Markov property. Briefly explain why each matters.

Solution

Possible answers include **fuel remaining** and **battery charge**. Fuel dictates how many future Thrust actions are possible, while battery charge dictates whether the satellite will be forced into a Sleep state soon. Both directly affect the probability of future transitions and rewards independently of the history of how the satellite reached its current slot.

- (b) Give one realistic reason a Thrust action could have more than one outcome. How should the transition model adapt to this?

Solution

A Thrust action might fail to reach the next slot due to a thruster misfire, micro-debris impact, or solar wind drag. The transition model must assign probability distributions to these outcomes (e.g., $P(\text{Move}) = 0.9, P(\text{Stay}) = 0.1$) rather than modeling the action as strictly deterministic.

Practice 2: Policies, Values, and Return

An automated supply convoy follows the deterministic fixed policy below. Use a discount factor of $\gamma = 0.25$.

State	Policy Action	Immediate Reward	Next State
S_A	Drive	+3	S_B
S_B	Unload	+8	Terminal

- (a) Starting in state S_A and following the policy, compute the discounted return.

Solution

$$\begin{aligned}\text{Return} &= r_1 + \gamma(r_2) \\ &= 3 + 0.25(8) \\ &= 3 + 2 \\ &= \mathbf{5}\end{aligned}$$

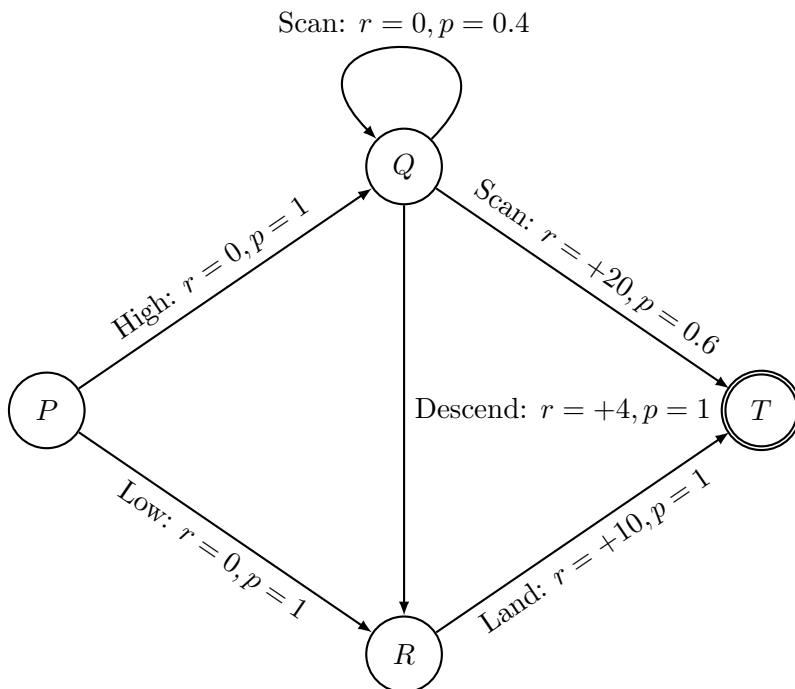
- (b) Suppose a different policy chooses “Airdrop” at S_A , receiving an immediate reward of +6 and terminating. Which policy gives the larger expected return from S_A ?

Solution

The original policy yields a return of **5**. The alternative policy yields a return of **6**. Therefore, the **alternative policy** (Airdrop) gives the larger expected return from S_A .

Practice 3: Value Iteration

An autonomous UAV navigates mountainous terrain. Let $\gamma = 0.5$, $V(\text{Terminal}) = 0$, and $V_0(P) = V_0(Q) = V_0(R) = 0$. Use synchronous updates.



- (a) Complete the first Value Iteration sweep to find $V_1(P)$, $V_1(Q)$, and $V_1(R)$.

Solution

For Q , the stochastic Scan action yields $0.6(20) + 0.4(0) = 12$. The Descend action yields $4 + 0.5(0) = 4$.

$$V_1(Q) = \max(12, 4) = \mathbf{12}$$

For R , Land yields $10 + 0.5(0) = 10$.

$$V_1(R) = \mathbf{10}$$

For P , High yields $0 + 0.5(0) = 0$. Low yields $0 + 0.5(0) = 0$.

$$V_1(P) = \max(0, 0) = \mathbf{0}$$

- (b) Using V_1 , compute only $V_2(P)$. Show the comparison between the available actions.

Solution

$$\text{High} = 0 + 0.5(V_1(Q)) = 0 + 0.5(12) = 6$$

$$\text{Low} = 0 + 0.5(V_1(R)) = 0 + 0.5(10) = 5$$

$$V_2(P) = \max(6, 5) = \mathbf{6}$$

- (c) If the convergence tolerance is $\theta = 0.1$, and the agent calculates $V_k(P) = 5.20$ and $V_{k+1}(P) = 5.26$ (with no other states changing), has Value Iteration converged?

Solution

Yes. The maximum change across the sweep is:

$$\begin{aligned}\Delta &= |5.26 - 5.20| \\ &= 0.06\end{aligned}$$

Since $0.06 < 0.1$, the algorithm has satisfied the convergence threshold.

Practice 4: Policy Iteration

An Electronic Warfare AI selects jamming frequencies. For the current policy, $V^\pi(S_C) = 12$ and $V^\pi(S_D) = -8$. Use a discount factor of $\gamma = 0.5$.

State	Action	Next State	Reward
S_A	Act 1	S_C	-1
S_A	Act 2	S_D	-1
S_B	Act 3	S_C	-1
S_B	Act 4	S_D	-1

The current policy selects **Act 2** at S_A and **Act 4** at S_B .

- (a) Under the current policy, compute $V^\pi(S_A)$.

Solution

The current policy selects Act 2 (transitioning to S_D).

$$\begin{aligned}V^\pi(S_A) &= -1 + 0.5(V^\pi(S_D)) \\ &= -1 + 0.5(-8) \\ &= -1 - 4 \\ &= \mathbf{-5}\end{aligned}$$

- (b) Perform policy improvement at state S_A . Compare the available actions and give $\pi'(S_A)$.

Solution

$$\text{Act 1 (to } S_C) = -1 + 0.5(12) = -1 + 6 = 5$$

$$\text{Act 2 (to } S_D) = -1 + 0.5(-8) = -1 - 4 = -5$$

Because Act 1 yields a much higher value ($5 > -5$), the improved policy updates to $\pi'(S_A) = \mathbf{Act\ 1}$.

Practice 5: Q-Learning from Experience

An autonomous rover has the following Q-values: $Q(R_1, \text{Drive}) = 0$, $Q(R_2, \text{Sample}) = 0$, $Q(R_2, \text{Wait}) = 4$. Let $\alpha = 0.5$, $\gamma = 0.5$, and let T be terminal.

- (a) The agent experiences $(R_2, \text{Sample}, 10, T)$. Compute the updated value of $Q(R_2, \text{Sample})$.

Solution

Since T is terminal, $\max_{a'} Q(T, a') = 0$.

$$\begin{aligned} Q(R_2, \text{Sample}) &= Q(R_2, \text{Sample}) + \alpha[r + \gamma(0) - Q(R_2, \text{Sample})] \\ &= 0 + 0.5[10 + 0 - 0] \\ &= \mathbf{5} \end{aligned}$$

- (b) The agent immediately experiences $(R_1, \text{Drive}, 0, R_2)$. Compute the updated value of $Q(R_1, \text{Drive})$.

Solution

After part (a), the maximum value at state R_2 is $\max(5, 4) = 5$.

$$\begin{aligned} Q(R_1, \text{Drive}) &= Q(R_1, \text{Drive}) + \alpha[r + \gamma \max_{a'} Q(R_2, a') - Q(R_1, \text{Drive})] \\ &= 0 + 0.5[0 + 0.5(5) - 0] \\ &= 0.5[2.5] \\ &= \mathbf{1.25} \end{aligned}$$